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Motion Anlaysis in Lower Extremity Joints during Ski Carving Turns Using Wearble Inertial Sensors and Plantar Pressure Sensors

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***Abstract*— Skiing is one of the most popular winter sports in the world. Even though the equipment has been improved for preventing injuries during skiing, injury risks of the lower extremity are still high. It is necessary to investigate injury risk by motion analysis during skiing. The wearable motion capture system consisting of inertial sensors and insole pressure sensors can be utilized due to restrictions of conventional optical motion capture system. In this study, the motions for short- and middle-turns during skiing were analyzed using a wearable motion capture system and the multi-scale computer simulation technology. Seven male certified ski coaches participated in this study and their full body motion and foot pressure data were simultaneously recorded by the wearable motion capture system. Joint kinematics and kinetics in the hip, knee and ankle of the right lower extremity were analyzed during short- and middle-turns. Even though a slight difference in the joint kinematics between two turns was predicted, the joint forces and moments for middle-turn were higher than those for short-turn. Because these higher joint forces and moments can result in osteoarthritis or ligament injury at the joint, injury risk of the joint for middle-turn may be higher than that for short-turn. This study confirms that the wearable motion capture system is useful for measuring the motion and plantar pressure data in outdoor sports such as skiing.**

Keywords—skiing; motion analysis; wearable motion capture system

I. INTRODUCTION

Skiing is one of the most popular winter sports in the world and there are approximately 400 million skiers in global. According to the US consumer product safety commission, the number of people treated for winter sports-related injuries in 2014 is more than 290,000 and the biggest amount of injuries (114,000 injuries) are from snow skiing. Even though the equipment, such as helmets, poles, bindings, and jackets, have been improved for preventing injuries during skiing, injury risks of the lower extremity are still high (40-64% of overall injuries during skiing) [1, 2]. Especially, knee injuries are the most common problem in professional skiers [3]. In order to prevent the injuries, it is necessary to investigate injury risk in various motions during skiing. Examining the joint kinematics and kinetics such as joint angle, angular velocity, angular acceleration, force and moment can be the important part of clinical diagnosis, planning of treatment, and evaluation of outcome.

Previous studies have investigated how skiing motions affect lower extremities for understanding and preventing injuries, using optical cameras and customized force plates [4,5]. Conventionally, motion analyses were performed using optical marker based motion capture system and ground mounted force plates, because the optical marker based motion capture system synchronized with force plates is well known for the high accuracy. However, this system also have some disadvantages such as restricted capturing area, limited foot placement and tied to the laboratory. In addition, this kind of motion analysis system is only available in few medical centers and research laboratories. Accordingly, it is difficult to readily analyze the motions of outdoor sports and activities, such as skiing. To solve these aforementioned, in this paper, we propose to use the wearable motion capture system consisting of inertial and magnetic measurement system and insole pressure system, which is a good alternative to the conventional motion capture system [6]. We also employed the multi-scale computer simulation technology for the human musculoskeletal system to analyze the skiing motions during short- and middle-turns.

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Fig. 1. Kinematic results (flexion-extension, abduction-adduction, internal-external rotation) in hip, knee, and ankle joints

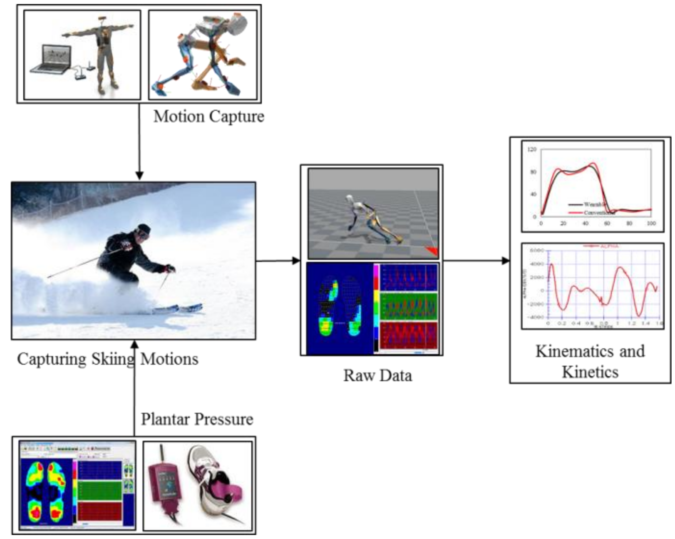


Fig. 2. Whole process to obtain kinematics and kinetics of lower extremity joints from motion capture and plantar pressure data

II. Methods

A. Experiments

Seven male certified ski coaches (age: 35.3 ± 4.9 yrs, height: 177.9 ± 1.3 cm, weight: 85.9 ± 3.0 kg, career: 15.0 ± 4.5 yrs) participated in this study. This work was approved by our institutional review board and the subjects were asked to give informed consent prior to experimentation. All subjects used their own equipment after taking the wearable motion capture system. The wearable motion capture system consisted of the MVN motion capture system (Xsens Technologies, Enschede, Netherlands) with 17 inertial sensors and the Pedar-X in-shoe foot pressure measurement system (Novel GmbH, Munich, Germany) of 85-99 pressure sensors with a measurable range of 15-600 kPa and resolution of 2.5 kPa [6]. The inertial sensors were attached to an MVN suit which the subjects were wearing and the pressure sensors were inserted in the ski boots. The power banks and signal transmitters of the both systems were fastened on the back of the subject (Fig. 1).

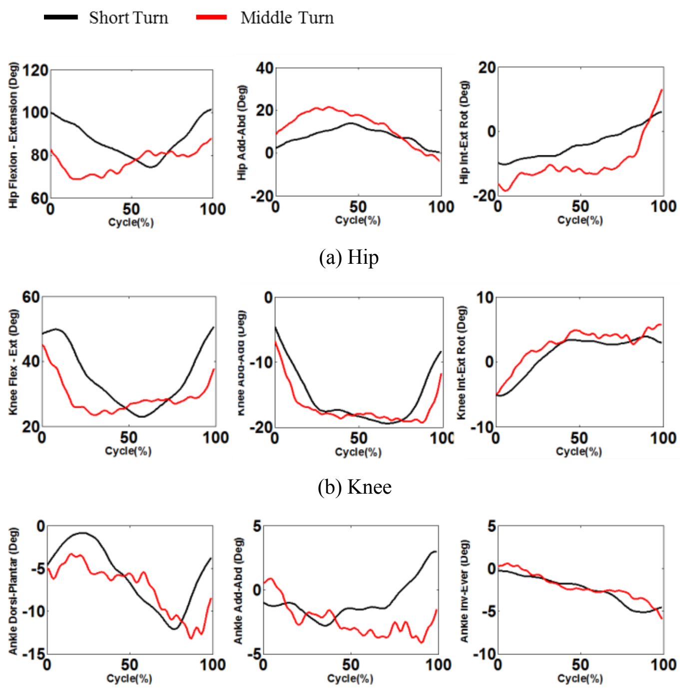
The skiing experiment was conducted in a ski slope with an angle of 17°-19° and a length of 570 m. During skiing, subjects performed both short- and middle-turns. A static calibration of the MVN system (referred as N-pose) and foot unloading calibration of the Pedar-X system were performed after warm-up. Body anthropometry data including body height, foot size, arm span, ankle height, hip height, hip width, knee height, and shoulder width were measured and used in the calibration process [7].

For each subject, the full body motion data, foot pressure, and the center of pressure (COP) data were simultaneously recorded with a sampling rate of 100 Hz. The motion data were exported from the MVN software as 3D positions of 64 anatomical landmarks. The GRF acting on the foot was predicted using measured foot pressure and COP data under the assumption that the resultant GRF was directed from the COP point to the center of gravity of the whole body [6]. The motion data and the GRF data were synchronized manually by matching the vertical motion of the foot and GRF data using Cortex software (Motion analysis, USA).

B. Data Analysis

Joint kinematics and kinetics of the right lower extremity were analyzed during left turns using short- and middle-turns because skiers use the opposite legs of the rotational directions. Here, the cross point between center of mass (COM) and ski's mean position was determined as start and end points of the turn [8]. The turns were normalized to 100% from the determined start point to the end point. The values of joint kinematics (angles) and kinetics (forces and moments) in the hip, knee and ankle were then calculated by the conventional inverse dynamics technique [6]. Fig. 2 shows the whole process to obtain the kinematics and kinetics of lower extremity joints from motion capture and plantar pressure data. A previously validated model of the human body developed in Matlab® (MathWorks, Natick, MA, USA) was used in the study to calculate three-dimensional (3D) kinematics and kinetics of the knee joint [6,9].

A dynamic model of the human whole body, consisting of 16 segments (head, thorax, lumbar, pelvis, upper arms, forearms, hands, thighs, shanks, and feet) and 45 degrees of freedom linkages for 15 joints (cervical, thoracic, lumbar, shoulders, elbows, wrists, hips, knees, and ankles), was utilized as in our previous study [9]. Joint centers and reference frames of each segment were defined based on previous studies [10,11]. The joint angles were calculated as the Euler angles of the distal segment reference frame relative to the proximal segment reference frame. Joint kinematic values, such as the angular velocity and acceleration of each segment, were calculated from motion capture data by the finite difference technique. The magnitudes of the joint forces and moments were then calculated based on a standard inverse dynamics analysis using the motion data and GRF [6,9,12]. For each



(a) Hip

(b) Knee

(c) Ankle

Fig. 3. Joint angles in hip, knee, and ankle joints in flexion (Flex,+) - extension (Ext,-), abduction (Abd,+) - adduction (Add,-), and internal (Int,+) - external (Ext,-) rotations

subject, the 3D kinematics and kinetics of the knee joint were quantified in the short and middle turns. The results of all subjects were averaged and plotted over 100 steps. The knee joint kinetics, such as joint forces and moments, were normalized to the respective body mass.

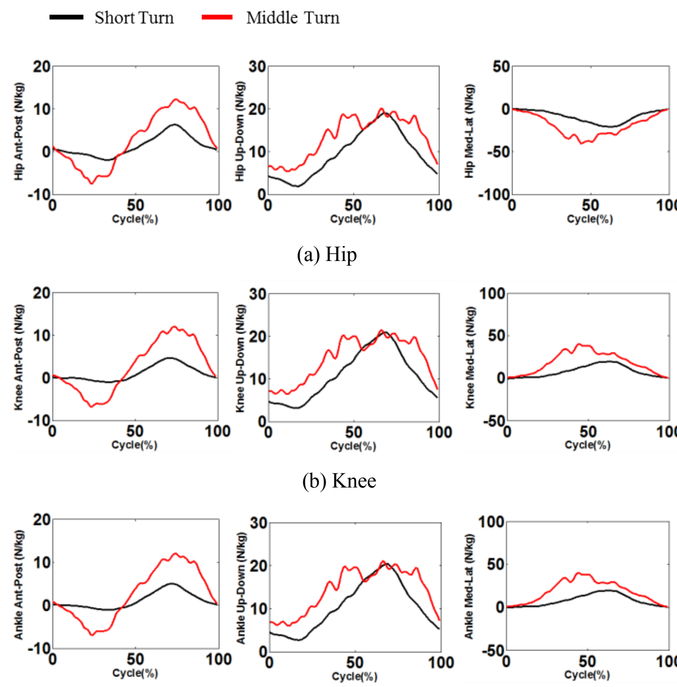
III. RESULTS

A. Joint Angles

The ranges of motion for short- and middle-turns were 27.1° and 19.0° in flexion-extension, 13.6° and 25.3° in abduction-adduction, and 16.4° and 31.6° in internal and external rotation, respectively, in hip joint (Fig. 3). Those in knee joint were 27.8° and 21.8° in flexion-extension, 14.9° and 12.6° in abduction-adduction, and 9.1° and 10.7° in internal and external rotation, respectively, and those in ankle joint were 11.2° and 9.9° in flexion-extension, 4.9° and 6.5° in abduction-adduction, and 5.8° and 5.0° in internal and external rotation, respectively (Fig. 3). The joint angle trends in short- and middle-turns were generally similar. For middle-turn, hip and knee joints showed faster extension motions than those for short-turn.

B. Joint Forces

The maximum joint forces for short- and middle-turns were 6.3 and 12.3 N/kg in anterior direction, 19.1 and 20.2 N/kg in up direction, and 20.6 and 40.1 N/kg in medial direction, respectively, in hip joint (Fig. 4). Those in knee joint were 4.7 and 12.1 N/kg in anterior direction, 21.0 and 21.5 N/kg in up

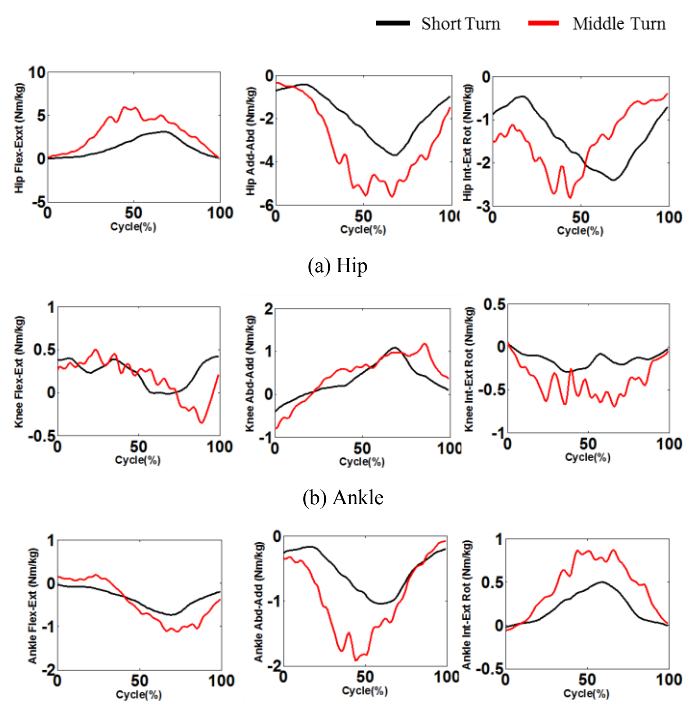


(a) Hip

(b) Knee

(c) Ankle

Fig. 4. Joint forces in hip, knee, and ankle joints in anterior (Ant, +) - posterior (Post, -), up (Up, +) - down (Down,-), and lateral (Lat,+) – medial (Med,-) directions



(a) Hip

(b) Ankle

(c) Ankle

Fig. 5. Joint moments in hip, knee, and ankle joints in flexion (Flex,+) - extension (Ext,-), abduction (Abd,+) - adduction (Add,-), and internal (Int,+) - external (Ext,-) rotation axes

direction, and 19.6 and 40.3 N/kg in lateral direction, respectively, and those in ankle joint were 5.1 and 12.1 N/kg in anterior direction, 20.4 and 21.1 N/kg in up direction, and 19.6 and 40.3 N/kg in medial direction, respectively (Fig. 4). Joint forces at each joint in all directions for middle-turn were higher than those for short-turn. Especially, the maximum knee joint forces in anterior and posterior directions for middle-turn were 2.4 times and 6.8 times higher than those for short-turn.

C. Joint Moments

The maximum joint moments for short- and middle-turns were 3.1 and 6.0 Nm/kg in flexion, 3.7 and 5.6 Nm/kg in adduction, and 2.4 and 2.8 Nm/kg in external rotation, respectively, in hip joint (Fig. 5). Those in knee joint were 0.4 and 0.5 Nm/kg in flexion, 1.1 and 1.2 Nm/kg in abduction, and 0.3 and 0.7 Nm/kg in external rotation, respectively, and those in ankle joint were 0.7 and 1.1 Nm/kg in extension, 1.0 and 1.9 Nm/kg in adduction, and 0.5 and 0.9 Nm/kg in internal rotation, respectively (Fig. 5). Joint moments at each joint in all directions for middle-turn were also higher than those for short-turn. Especially, the maximum knee joint moment in internal rotation for middle-turn was 2.4 times higher than that for short-turn

IV. DISCUSSION

The hip and knee flexion-extension angles showed different patterns between short- and middle-turns. The mean joint flexion angles in hip and knee joints for middle-turn were approximately 10° and 6° less than those for short-turn, respectively. However, the patterns in abduction-adduction and internal-external rotation angles were quite similar in two turns. In addition, the ranges of motion in all rotation directions were also similar in two turns. Higher joint forces and moments were predicted in all joints. Especially, the knee joint force in anterior-posterior direction and the knee joint internal moments were higher for middle-turn than for short-turn. Higher joint forces and moments for middle-turn may have a possibility to result in joint injuries or diseases such as osteoarthritis. Thus, it is recommended to use proper turns for the ski player in the rehabilitation process from the knee joint injury as a safer turn. In general, a radius of rotation during the short-turn motion is smaller than that during the middle-turn motion, and the smaller radius may generate higher centrifugal force which can produce higher joint forces and moments in lower extremity.

V. CONCLUSIONS

Previous studies have used optical cameras and customized force plates for measuring motion and ground reaction force, but they could obtain the data in restricted spaces during skiing. In this study, motion and pressure data of skiers could be measured using a wearable motion capture system Joint kinematics and kinetics could be calculated using both measured motion and plantar pressure data. Even though slight difference of joint kinematics between short-turn and middle-turn were predicted, joint forces and moments for middle-turn were higher compared to those for short-turn. Because these higher joint forces and moments can result in osteoarthritis and ligament injury at the joint, injury risk of the joint for middle-turn may be higher than that for short-turn on the aspect of the osteoarthritis and ligament injuries. This study confirms that the wearable motion capture system is useful for measuring the motion and plantar pressure data during outdoor sports such as skiing.

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